# Forward Propulsion Using Ionic Thruster

Group B3

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## INTRODUCTION

Ionic thrusters generate thrust by accelerating charged particles using a strong electric field, a principle known as electrohydrodynamic (EHD) propulsion[1][2]. Unlike conventional propulsion systems, they have no moving mechanical parts, operate with very low noise, and can achieve high efficiency at small scales[3]. These characteristics make ionic propulsion a promising technology for lightweight aerial platforms such as gliders and micro-UAVs.

The practical development of ion propulsion was significantly advanced by NASA with the successful in-space test of the SERT-1 mission in 1964. Since then, ion thrusters have powered deep-space missions such as Deep Space 1 and Dawn, demonstrating exceptional fuel efficiency and long operational lifespan.

Motivated by the growing need for quieter and more environmentally friendly propulsion technologies, this project focuses on designing a compact laboratory-scale ionic thruster capable of generating measurable thrust. The goal is to explore its feasibility for lightweight glider propulsion and investigate the engineering challenges of adapting ion propulsion for atmospheric application

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## OBJECTIVES

- Design a compact ionic thruster.
- Optimize electrode geometry and spacing for maximum thrust
- Fabricate a working prototype using selected materials
- Experimentally evaluate thrust performance and feasibility for glider use

## COMPLEX ENGG. PROBLEM

**Electrical Challenges**

- Reverse ion flow and charge recombination reducing thrust *(WP3, WK3)*
- Electrical discharges and corona instability at high voltage *(WP1, WK4)*
- Transformer and high-voltage component failures *(WK6)*
- Corona losses reducing propulsion efficiency *(WP2)*

**Flow & Performance Challenges**

- Non-uniform ion distribution affecting thrust generation *(WP3, WK2)*
- Instability in ion flow behavior under atmospheric conditions *(WP4)*
- Limited thrust-to-power ratio requiring optimization *(WP1)*

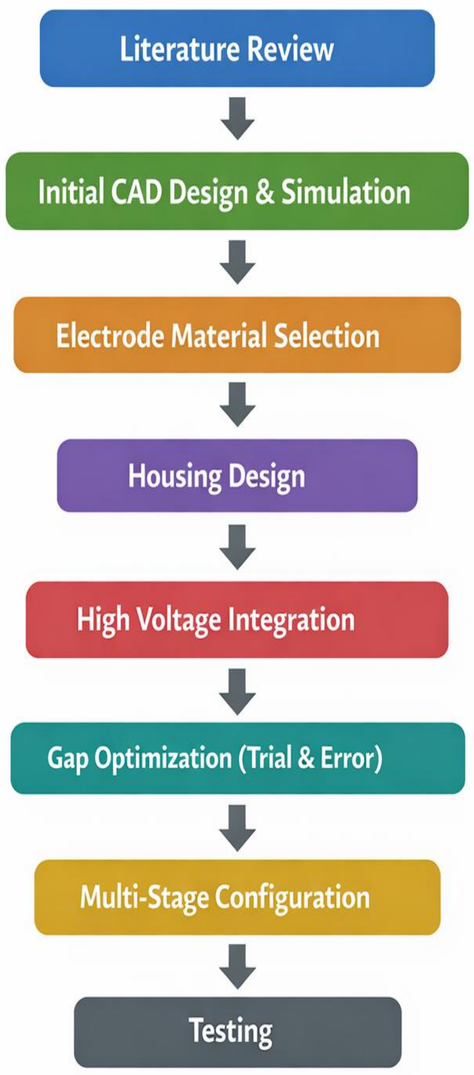
**Mechanical & Structural Challenges**

- Precision electrode alignment required for stable electric field *(WP7)*
- Lightweight yet electrically insulated structure design *(WP2, WK5)*

**Engineering Activities (EA)**

- Integration of high-voltage system, electrode design, and structural housing *(EA1, EA2)*
- Experimental optimization of electrode spacing and multi-stage configuration *(EA3)*
- Prototype testing and performance evaluation of ionic propulsion *(EA5)*

## METHODOLOGY



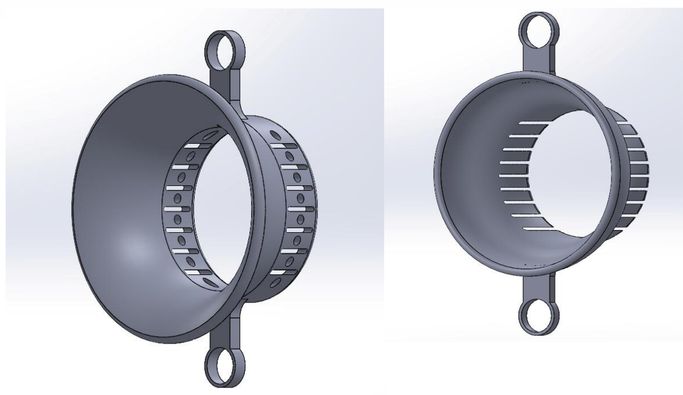



*Fig 1 . CAD design*

*Fig 2 . Final Setup*

## RESULTS & DISCUSSION

The ionic thruster successfully generated measurable airflow, confirming effective ionization and thrust production in the laboratory-scale prototype. Stable ionic wind was observed under optimal electrode spacing and alignment, demonstrating successful implementation of the working principle.

Occasional sparking and audible discharge were noticed during operation, indicating localized electric field concentration and minor corona instability, likely due to edge irregularities or slight misalignment. Despite this, overall airflow performance remained stable.

Although designed for approximately 40 kV operation, the actual voltage could not be consistently verified and was likely lower, limiting maximum thrust output. The step-up transformer also experienced durability issues after repeated use.

Overall, the prototype achieved reasonable thrust for its size. Future improvements should focus on precise voltage regulation, improved transformer robustness, and precision-fabricated electrodes to enhance field uniformity and reduce sparking, enabling more rigorous performance evaluation.

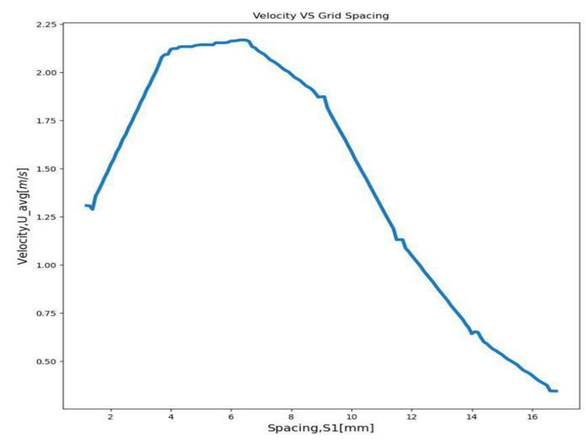



*Fig 3. Velocity VS Spacing*

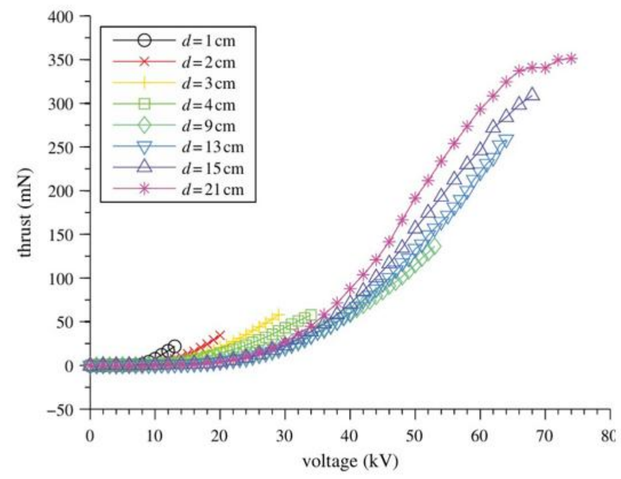



*Fig 4 . Thrust VS Voltage*

## REFERENCES

1. D. M. Goebel, I. Katz, and I. G. Mikellides, Fundamentals of Electric Propulsion, 2nd ed. Hoboken, NJ, USA: Wiley, 2023.
2. V. Y. Khomich, V. E. Malanichev, and I. E. Rebrov, "Electrohydrodynamic thruster for near-space applications," Acta Astronautica, vol. 180, pp. 141–148, 2021, doi: 10.1016/j.actaastro.2020.12.002.
3. E. Stuhlinger, Ion Propulsion for Space Flight. New York, NY, USA: McGraw-Hill, 1964